

Trainings ▾

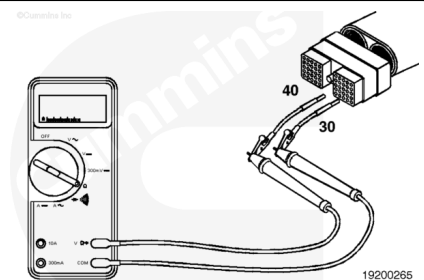
Resistance Check

Disconnect the actuator harness connector from the ECM.
Make sure the shaft speed sensor is connected to the OEM harness.

⚠ CAUTION ⚠

Do not use probes or leads other than Part Number 3822758. The OEM harness will be damaged. The leads must fit tightly in the connector without expanding the pins in the connector.

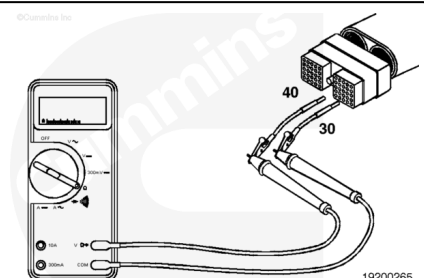
Insert a test lead into pin 40 of the actuator harness connector.
Insert the other lead into pin 30 of the connector.



Connect the two alligator clips to the two probes of the multimeter. Adjust the multimeter to the resistance setting and measure resistance. The resistance value **must** be 750 to 1500 ohms. If the value is **not** correct, there is a problem with the actuator harness, provided that the shaft speed sensor has been previously checked.

Repair or replace the actuator harness or the OEM harness, depending on the location of the damage. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/82/82-019-043-tr.html).

If the value is correct, the circuit **must** still be checked for a short circuit to ground and a short circuit from pin to pin.



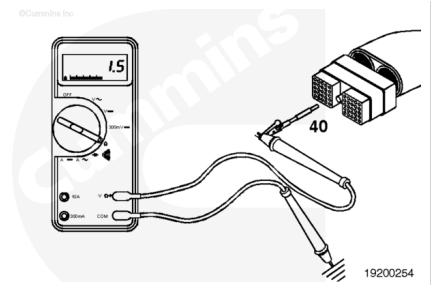
Check for Short Circuit to Ground

Check for a short circuit to ground. Insert the multimeter probe with attached test lead into pin 40 of the actuator harness connector. Touch the other multimeter probe to the engine

block. Measure the resistance. The multimeter **must** show an open circuit (100k ohms or more).

If the circuit is **not** open, there is a short circuit to ground in the shaft speed sensor circuit in the actuator harness or OEM harness.

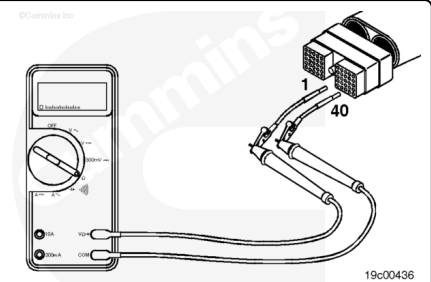
Repair the wires which are shorted in the circuit according to the vehicle manufacturer's procedures.



Check for Short Circuit from Pin to Pin

Check for a short circuit from pin to pin. Insert the multimeter probe with attached test lead into pin 40 of the actuator harness connector. Insert the other test lead into pin 1 of the connector.

Measure the resistance. The multimeter **must** show an open circuit (more than 100k ohms).



Remove the test lead from pin 1 and test all pins in the connector.

Measure the resistance from pin 40 of the actuator harness connector to all other pins in the connector, one at a time. The multimeter **must** show an open circuit (more than 100k ohms) at all pins.

Note : If the values are correct for all of the circuit checks, the shaft speed sensor circuit is good.

Connect all components after completing the repair.

